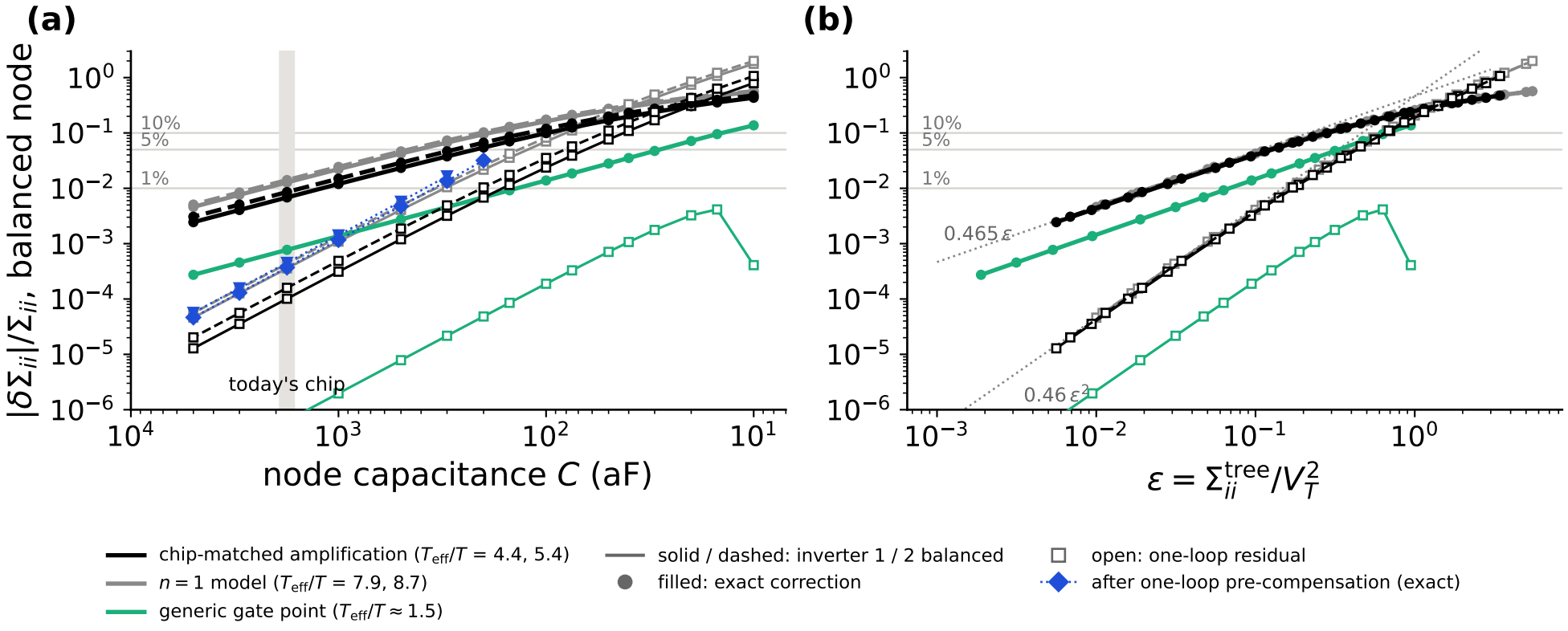
(a)
(b)
$|\delta\Sigma_{ii}|/\Sigma_{ii}$, balanced node
10%
5%
1%
today's chip
node capacitance C (aF)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0.465 ε
0.46 ε^2
$\varepsilon = \Sigma_{ii}^{\mathrm{tree}}/V_T^2$
chip-matched amplification (T_{eff}/T = 4.4, 5.4)
$n = 1$ model (T_{eff}/T = 7.9, 8.7)
generic gate point ($T_{\mathrm{eff}}/T \approx 1.5$)
solid / dashed: inverter 1 / 2 balanced
filled: exact correction
open: one-loop residual
after one-loop pre-compensation (exact)